ATELIER LOGIC GROUP LLC

DBA RELULAYER™ • ADVANCED R&D & HIGH-ASSURANCE SYSTEMS

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UEI: G9WFN3AL69J3
CAGE: 20JX8
NAICS: 541715
PSC: AC11
SAM.GOV ACTIVE ENTITY

CORPORATE OVERVIEW

Atelier Logic Group LLC is a Louisiana-based advanced R&D firm specializing in deterministic safety micro-kernels, artificial intelligence governance, and mathematical optimization for defense, aerospace, and autonomous systems. Our flagship product, **ReluLayer™**, is a sub-microsecond hardware-level safety governor that interdicts unsafe autonomous commands with certifiable, memory-safe execution.

128 Cycles	267 ns	0.00%	326 B
DETERMINISTIC WCET	WORST-CASE EXECUTION TIME	HIL VARIANCE (5,000 PASSES)	CORE KERNEL FOOTPRINT

CORE CAPABILITIES

- **CISA Memory-Safe Rust Kernel:** Bare-metal `#![no_std]` safety micro-kernel with zero dynamic heap allocation (guaranteed zero alloc/heap dependency).
- **Contested C2 & EW Kinematic Protection:** Bare-metal `#![no_std]` Rust kernel operates on-chip to maintain deterministic flight envelope enforcement during total RF jamming, SATCOM loss, or C2 link degradation.
- **Deterministic Interdiction:** 128 cycles / 267 ns worst-case execution time at 480 MHz on ARM Cortex-M7 / STM32H7 class targets.
- **Certifiable Process Alignment:** RTCA DO-178C DAL A / MIL-STD-882E safety-critical engineering and hazard analysis workflows.
- **Operations Research & MILP:** Large-scale optimization engines for fleet, gate, and logistics scheduling.
- **Hardware-in-the-Loop (HIL):** Monte Carlo stress testing and digital-twin validation in Kenner, LA.

DIFFERENTIATORS & IP

- **Zero-RF Air-Gapped Enforcement:** Unlike cloud-tethered or network-dependent AI safety layers, ReluLayer™ operates entirely on local edge silicon in 128 CPU cycles (267 ns). It prevents kinetic stalls or erratic control commands when electronic warfare severs satellite (e.g., Starlink/GPS) connections.
- **Patent Pending:** USPTO Serial No. 64/127,611 — ACCSE™ 2-Tier Decentralized Asymmetric Swarm Interdiction (Attorney Docket RELU-001-PPA7, filed August 7, 2026); portfolio continuity from Provisionals I–VI.
- **HIL-Proven Reliability:** Zero violation rate across 5,000 hardware-in-the-loop stress passes.
- **Enterprise AI Leadership:** 30+ cumulative years of enterprise AI & engineering leadership, including DXC Technology's 10,000-member global AI Guild.
- **100% On-Shore:** Engineered, tested, and maintained in Kenner, Louisiana.
- **Transition Ready:** Phase I / Open Topic Direct-to-Phase II Candidate | TRL-4 Dual-Use Technology for Army DEVCOM, PEO Aviation, NAVAIR, AFWERX, and DARPA.

DESIGNATED TACTICAL USE CASES

- **Autonomous UAS Operations in Jammed/Denied Theaters:** Guarantees structural and aerodynamic stability during fallback transitions when primary guidance feeds are jammed, spoofed, or blinded by high-frequency EW systems.
- **Flight-Critical Safety Interdiction:** Hard real-time envelope enforcement for fixed-wing, rotary-wing, and vertical-lift autonomy stacks.
- **Legacy C/C++ Migration for Safety-Critical Loops:** Refactoring of flight control, navigation, and sensor-fusion code to certifiable `#![no_std]` Rust.
- **MILP-Based Fleet & Logistics Optimization:** Resource scheduling under contested or degraded supply-chain conditions.

NAICS & PSC CODES

Primary NAICS: 541715 (Research & Development in the Physical, Engineering, and Life Sciences)

Secondary NAICS: 541330, 541512, 541690

Key PSCs: AC11, AC12, AJ12, AJ13, DA01, D302

ENGAGEMENT MODELS

Prime / Subcontractor: Defense R&D, Direct-to-Phase II SBIR/STTR, and OTA Alignment.

OEM Licensing: ReluLayer™ integration for flight-stack and autonomy platforms.

Custom R&D: Tailored high-assurance algorithms and safety certification support.

PRINCIPAL INVESTIGATOR

Ishmael Ibrahim Amin, M.S. — Principal Investigator & Founder

Northwestern University M.S. Data Science in Artificial Intelligence • UC Santa Barbara B.S. Statistical Science. Former leader of DXC Technology's 10,000-member AI Guild, with a track record of deploying enterprise-scale machine learning and delivering $20M–$40M in operational efficiency outcomes.

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